

Leveraging Input-Level Feature Deformation with Guided-Attention for Sulcal Labeling

Seungeun Lee, Seunghwan Lee, Ethan H. Willbrand, Benjamin J. Parker, Silvia A. Bunge, Kevin S. Weiner, and Ilwoo Lyu

Abstract—The identification of cortical sulci is key for understanding functional and structural development of the cortex. While large, consistent sulci (or primary/secondary sulci) receive significant attention in most studies, the exploration of smaller and more variable sulci (or putative tertiary sulci) remains relatively under-investigated. Despite its importance, automatic labeling of cortical sulci is challenging due to (1) the presence of substantial anatomical variability, (2) the relatively small size of the regions of interest (ROIs) compared to unlabeled regions, and (3) the scarcity of annotated labels. In this paper, we propose a novel end-to-end learning framework using a spherical convolutional neural network (CNN). Specifically, the proposed method learns to effectively warp geometric features in a direction that facilitates the labeling of sulci while mitigating the impact of anatomical variability. Moreover, we introduce a guided-attention mechanism that takes into account the extent of deformation induced by the learned warping. This extracts discriminative features that emphasize sulcal ROIs, while suppressing irrelevant information of unlabeled regions. In the experiments, we evaluate the proposed method on 8 sulci of the posterior medial cortex. Our method outperforms existing methods particularly in the putative tertiary sulci. The code is publicly available at <https://github.com/Shape-Lab/DSPHARM-Net>.

Index Terms—Sulcal Labeling, Spherical CNN, Spherical Warp, Guided-Attention

I. INTRODUCTION

The posterior medial cortex (PMC) is considered a central hub of the default mode network [1]–[3] and implicated in a

broad array of cognitive functions, such as episodic memory, self-referential processing, and cognitive control [4]–[6]. Thus, understanding the precise morphology of sulci has important implications for how individual differences in structural-functional relationships in PMC relate to individual differences in behavior and cognition. Interestingly, recent studies have uncovered new putative tertiary sulci including a new tripartite landmark in PMC and further shown that the sulcal patterning in human PMC is different compared to non-human hominoids [7]. Recent findings show that the morphology of a subset of PMC sulci is different between neurotypical controls and several clinical populations [7]–[9]. Therefore, automated tools to identify PMC sulci could have substantial translational applications. Coming closer to achieving this goal of accurate automatic definition of these neuroanatomical structures, the development of sulcal labeling tools continues to make steady advances [10]–[19]. Unfortunately, existing tools lack the ability to accurately label these fine-grained sulci, hindering progress in their study (see Fig. 1 for anatomical variability).

Convolutional neural networks (CNNs) have shown remarkable achievements and have emerged as a promising approach for cortical data processing [20]–[26]. However, applying CNNs to the sulcal labeling task presents significant challenges, primarily due to three main reasons: (1) the presence of substantial anatomical variability, (2) the class-imbalance problem that arises due to the significant disparity in the size of the regions of interest (ROIs) and unlabeled regions, and (3) the limited availability of manually annotated data. To the best of our knowledge, only a single study [23] has presented labeling of both primary/secondary and putative tertiary sulci. Despite achieving improved performance, their approach involves the use of spherical registration for data augmentation, which in turn relieves the data scarcity issue, whereas the class-imbalance problem remains largely unexplored.

In this paper, we propose a novel end-to-end learning framework that leverages a spherical CNN for labeling of sulci in PMC. Our driving motivation is to tackle both anatomical variability and class imbalance effectively. To this end, the proposed method is centered around two key components: (1) the learning of a spherical warp without a specific target, to minimize the influence of anatomical variability, and (2) the incorporation of a guided-attention mechanism that considers the extent of deformation induced by the learned warping to focus on sulcal ROIs while suppressing unnecessary information in unlabeled regions. The main contributions of this work

This work was supported by the NRF under RS-2023-00251298, RS-2023-00266120, and RS-2024-00333931, by the IITP AIGS Program under RS-2020-II201336 (UNIST) and RS-2019-II191906 (POSTECH), by NSF CAREER Award 2042251, and by the NIH MSTP under T32 GM140935. Pediatric neuroimaging data were provided by the NORA study funded by the NIH NINDS under R01NS057156 and the NSF under BCS1558585. Adult neuroimaging data were provided by the HCP, WU-Minn Consortium under Grant 1U54-MH-091657 funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research, and the McDonnell Center for Systems Neuroscience at Washington University. (Corresponding author: Ilwoo Lyu)

S. Lee is with AI Research Team, Kileon, Seoul 04637, South Korea. S. Lee and I. Lyu are with the Department of Computer Science and Engineering, POSTECH, Pohang 37673, South Korea.

E. Willbrand is with the School of Medicine and Public Health, The University of Wisconsin, Madison, WI 53726, USA.

B. Parker and K. Weiner are with the Helen Wills Neuroscience Institute, The University of California, Berkeley, CA 94720, USA.

S. Bunge and K. Weiner are with the Department of Psychology, The University of California, Berkeley, CA 94720, USA.

I. Lyu is with the Graduate School of Artificial Intelligence, POSTECH, Pohang 37673, South Korea. (email: ilwoolyu@postech.ac.kr)

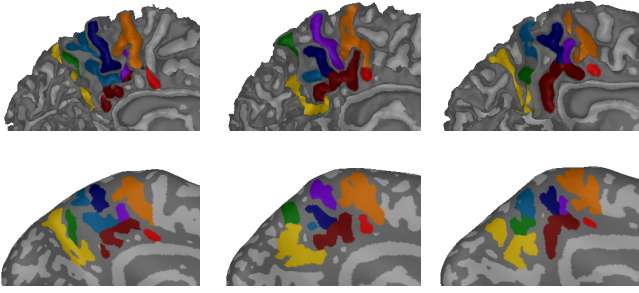


Fig. 1. Anatomical variability in PMC. Sulcal ROIs are overlaid on the white (top) and inflated surfaces (bottom). The remarkable anatomical variability of sulci is shown across individual participants. Even within the same sulcal category, there is significant anatomical difference.

can be summarized as follows:

- **Spherical warp** learns to amplify the scales of small sulcal regions and simplify their regional boundaries. This approach can reduce anatomical variability across individual participants and facilitate sulcal labeling.
- **Guided-attention mechanism** assigns weights (importance) to sulcal ROIs based on the extent of the learned warping. This approach can effectively discriminate sulcal ROIs from unlabeled regions.
- **Self-intersection loss** offers a high degree of flexibility in a spherical warp while preserving a diffeomorphic mapping. This approach ensures a smooth and invertible spherical warp without altering the spherical topology.
- **Comprehensive experiments** demonstrate the improved performance achieved by the proposed method in ablative, qualitative, and quantitative results.

II. RELATED WORK

A. Sulcal Curve Labeling

In previous attempts, sulcal labeling has relied on curves (skeletons) delineated along sulci such as [14], [15], [27]–[30]. For instance, in [13], a probabilistic graphical model was used for joint labeling of major sulcal curves through maximum a posteriori estimation. In [15], a discriminative model was employed, utilizing a boosting tree to extract major cortical sulci without the need for user-defined rules. In [16], a multi-atlas scheme was introduced for sulcal curve labeling to take into account sulcal variability. Their work was further extended to a learning-based framework in [18] by utilizing a 2D U-Net model. On another front, some studies proposed registration-based sulcal curve matching [17], [31]–[34]. However, these methods may encounter practical challenges, necessitating a well-defined prior model capable of fully encompassing sulcal variability or the acquisition of a sufficient number of references. Also, the methods are sensitive to the error-prone nature of sulcal curve extraction, which further adds to the complexity of the labeling task.

B. Sulcal Region Labeling

Unlike curve-based methods, other approaches can effectively delineate sulcal regions that span a wider area, providing

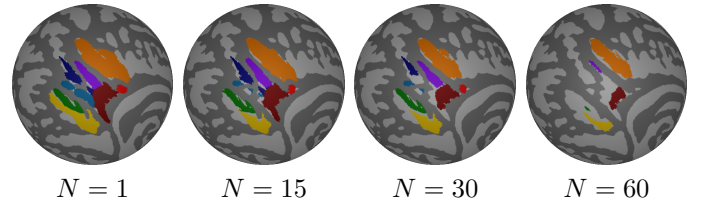


Fig. 2. Average sulcal patterns in PMC overlaid on *fsaverage* after surface registration. As more subjects are aligned, the detailed patterns become obscured due to the inherent anatomical variability that is not fully represented by the target patterns. The registration-like approach is thus limited in its ability for sulcal labeling, as also reported in [35].

a more comprehensive representation of sulcal patterns. They have primarily embraced two distinct aspects: (1) atlas-based labeling following surface registration [34], [36]–[38] and (2) sequential sulcal region segmentation and labeling [11], [12], [36], [39]–[42]. The BrainVisa pipeline [10] jointly achieves registration and labeling while fully utilizing the synergistic potential of both aspects. However, all these techniques suffer from template selection bias in surface registration, yielding misaligned sulcal patterns (see Fig. 2) and/or heavy computational demands, especially when processing large datasets or dealing with high-resolution cortical surfaces.

Only a few studies have explored the application of CNNs to sulcal labeling. [19] introduced a voxel-based sulcal labeling approach using a CNN. They used the BrainVISA pipeline [10] to preserve crucial geometrical and topological properties important for sulcus morphometry, effectively reducing spatial inconsistencies and inaccurate geometric definitions. Their approach offers a comprehensive set of sulcal ROIs; however, the identification of putative tertiary sulci was not thoroughly explored. Meanwhile, [23] introduced a novel approach that leverages context-aware learning based on neuroanatomical hierarchy and registration-based data augmentation to achieve sulcal labeling. To the best of our knowledge, this is the only study that achieves joint labeling of primary/secondary and putative tertiary sulci. Nevertheless, their approach requires a time-consuming sequential learning process, and the data augmentation requires meticulous selection of registration targets.

C. Spherical CNNs for Cortical Surface Segmentation

Remarkable progress has been made in the development of spherical CNNs designed specifically for cortical surface processing. [20] introduced a U-Net-based framework that operates on a semi-uniform spherical grid within the spherical domain. In [43], they further incorporated learnable filter offsets to support adaptive filter shapes for non-circular convolution. While promising, these studies depend on a specific spherical coordinate system, resulting in filter orientations being inverted at the poles. Moreover, the lack of rotational equivariance requires substantial data augmentation to fully cover $SO(3)$ configurations. Meanwhile, [44] introduced a spherical harmonics-based CNN. Their method enables the learning of high-frequency components in the spectral domain while maintaining rotational equivariance. They demonstrated improved performance in cortical parcellation without the need for data augmentation. While rotational equivariance in

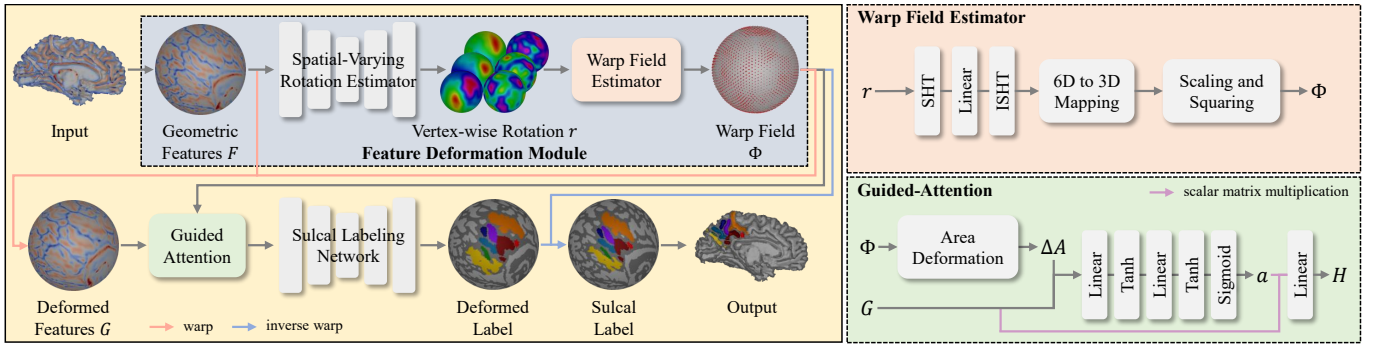


Fig. 3. A schematic overview of the proposed method. The geometric features F are mapped onto the unit sphere and used for the spatial-varying rotation parameter estimator. The estimated parameters r are transformed into the warp field Φ . Then, the deformed features G are obtained by applying Φ to F . The proposed guided-attention extracts the enhanced features H by assigning weights to regions based on the extent of deformation. H is passed through the sulcal labeling network and its output is recovered back to the original sphere by applying Φ^{-1} .

their method offers computational advantages by alleviating the need for extensive data augmentation, the learned filters are isotropic, which inherently restricts their flexibility in representation. This limitation can pose challenges, particularly in the context of sulcal labeling, where the presence of large anatomical variability necessitates more adaptable and nuanced representations of convolutional filters.

D. Small Region Segmentation in Computer Vision

The challenge of perceiving small ROIs has garnered significant interest particularly in the computer vision community, leading to the exploration of various approaches. Existing methods leverage context information [45], [46], scale-aware learning [47], [48], and attention mechanisms [49], [50]. Some approaches incorporate auxiliary networks [45], [46] that assess the relationship between objects and unlabeled regions surrounding small ROIs to leverage context information effectively. Scale-aware learning techniques such as fusion of multi-resolution features [47], [48] enhance the network's receptive field and representation capability. Attention mechanisms have shown potential for small region recognition [49], [50] by weighting small ROIs over background information. However, since these approaches are designed in Euclidean spaces, they might not yet be directly applicable to cortical surface data possessing unstructured grid nature.

III. METHODS

A. Problem Statement

Our goal is to label sulcal ROIs from a cortical surface $\Omega \subset \mathbb{R}^3$ using C distinct features of cortical geometry. Under the assumption of a genus-0 surface, valid spherical parameterization holds a one-to-one mapping of geometric features onto the unit sphere as a spherical function $F: \mathbb{S}^2 \rightarrow \mathbb{R}^C$. Here, sulcal labeling can be achieved by optimizing discriminant functions given the cortical geometry. At a spherical location $\vec{x} \in \mathbb{S}^2$, we infer its label $z \in \{1, 2, \dots, L\}$. Formally, we have an L -class non-linear discriminant function $\mathcal{F}: \mathbb{R}^C \rightarrow \mathbb{R}^L$ that takes F as an input data. The discriminant function $\mathcal{F}$ is optimized to provide the highest discriminant value for a given location, which determines the corresponding sulcal label. In the supervised learning, we train the learnable model $\mathcal{F}$ using

observations of the input features F and their corresponding known labels. Fig. 3 shows a schematic overview of our method. In the remainder of the section, we will describe the details of refining F and modeling $\mathcal{F}$.

B. Feature Deformation Module

As discussed, the substantial anatomical variability in sulci poses challenges for accurate sulcal labeling and necessitates a significant amount of data acquisition. To address this issue, we introduce a novel approach that involves learning deformation of the input geometric features. By incorporating these learned warps, we can alleviate the impact of anatomical variability and thereby facilitate the learning process. To this end, we find a diffeomorphic warp for each location. Inspired by our recent work [51], we model a warp field $\Phi: \mathbb{S}^2 \rightarrow \mathbb{S}^2$ as a stationary ordinary differential equation (ODE):

$$\frac{d\Phi(\vec{x}, t)}{dt} = v(\Phi(\vec{x}, t)), \quad (1)$$

where $v \in T_{\vec{x}}\mathbb{S}^2$ is a velocity at $\vec{x}$ with the initial condition $\Phi(\vec{x}, 0) = \vec{x}$. According to the ODE theories [52], the existence of unique trajectories over a velocity field depends on the Lipschitz continuity in $\vec{x}$. Because the solution becomes an exponential mapping, a spherical warp can be obtained by

$$\Phi(\vec{x}) = \lim_{n \rightarrow \infty} \Phi_{v/n} \circ \dots \circ \Phi_{v/n}(\vec{x}), \quad (2)$$

where $\Phi_v(\vec{x})$ is an exponential map $\exp_{\vec{x}}(v)$ at $\vec{x}$. Following [51], we consider a velocity on the sphere as a rotation:

$$\Phi_v(\vec{x}) = R_{\vec{x}} \cdot \vec{x}. \quad (3)$$

In [53], $R_{\vec{x}}$ can be decomposed into a 6D vector. Hence, the spatial-varying rotation is equivalent to a spherical function of 6D vectors. We denote such vertex-wise parametrization by $r: \mathbb{S}^2 \rightarrow \mathbb{R}^6$. We employ a spherical harmonics-based CNN (SPHARM-Net) [44] to model r because rotational data augmentation is entirely avoided by its rotational equivariance property and some resources can be shared for computational efficiency. Next, we generate a spatial-varying function r using the spherical harmonics transform (SHT) [51]:

$$\hat{c}_{lm} = \int_{\mathbb{S}^2} r(\theta, \phi) Y_{lm}(\theta, \phi) d\theta d\phi, \quad (4)$$

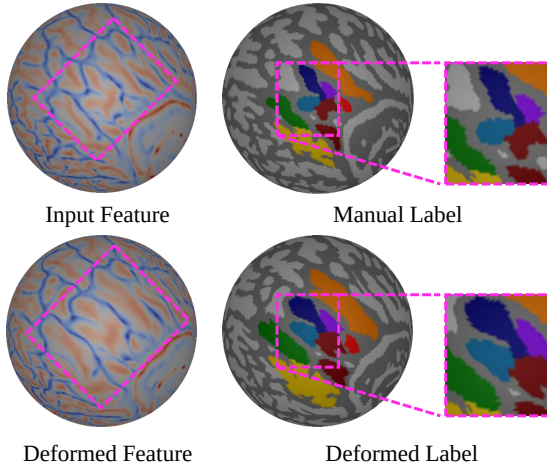


Fig. 4. Feature deformation. Cortical geometry (left): the expanded regions can be tessellated with larger sampling points. Sulcal labels (right): anatomical variability can be reduced by simplifying and expanding the boundaries of ROIs, which is learned during training. These effects contribute to facilitating sulcal labeling. The deformed regions are highlighted by pink dotted boxes.

where $Y_{lm} : \mathbb{S}^2 \rightarrow \mathbb{R}$ is the spherical harmonic basis function at degree of l and order m . To further refine the spectral components, we employ a parameter-wise linear model:

$$c_{lm} = \hat{c}_{lm} \cdot w_{lm} + b_{lm}, \quad (5)$$

where $w_{lm}, b_{lm} \in \mathbb{R}^6$ are learnable parameters. The smoothed function is then obtained via the inverse spherical harmonics transform (ISHT) by limiting its bandwidth B :

$$\bar{r}(\theta, \phi) = \sum_{l=0}^B \sum_{m=-l}^l c_{lm} Y_{lm}(\theta, \phi). \quad (6)$$

Finally, the refined parameter $\bar{r}$ is converted into a rotation matrix for each location via the Gram Schmidt process [53]. Because the velocity field is a collection of rotation matrices (Eq. (3)), the warp field can be computed as a composition of matrix multiplication. Therefore, for a stationary velocity field, we can efficiently solve Eq. (2) by matrix multiplication in a scaling and squaring fashion [51] with k -step recursion:

$$\Phi_{v/2^{k-1}}(\vec{x}) = \Phi_{v/2^k} \circ \Phi_{v/2^k}(\vec{x}). \quad (7)$$

The improved input features are obtained by $G = F \circ \Phi$. We observe that the proposed method simplifies sulcal patterns to enhance their representation; it expands the area of sulci while reducing structural complexity (*i.e.*, simplified boundaries), as shown in Fig. 4. It is noteworthy that the proposed method is not intentionally designed to scale up ROIs; rather, such a scaling effect is learned by the neural network during training. Also, unlike spherical registration, the proposed method does not require an explicit target, which grants greater flexibility of the spherical warp.

C. Guided-Attention Module

The majority of the cortical surface comprises unlabeled regions, but only a small portion of the regions are of interest for the labeling of specific sulci. This poses a significant

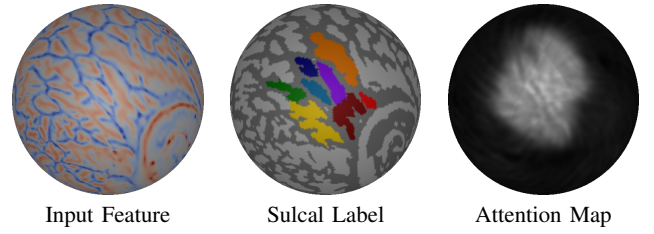


Fig. 5. An attention map generated by the proposed guided-attention module. The proposed module effectively enhances the input feature by directing its focus towards ROIs to be labeled. This assigns more weight to the expanded regions by the feature deformation module.

challenge in terms of class imbalance, where the uneven distribution of labeled and unlabeled regions across the whole cortex. Therefore, we employ an attention mechanism to focus on to-be-labeled regions. In our approach, regions with larger expansion play a crucial role as they receive more attention in the spherical warp, as illustrated in Fig. 5. This is due to the fact that, during backpropagation, regions with larger labeling errors necessitate a more extensive spherical warp. To this end, we measure area difference after deformation at each location from the warp field Φ to assign weights. Due to the limited availability of manual annotation, we adapt a scalar matrix multiplication-based attention mechanism [49], [54] reducing the number of learnable parameters to prevent overfitting.

To determine the importance of each location, we utilize the vertex-wise area difference before and after deformation. The vertex-wise area can be calculated by summing the areas of the adjacent triangles. For each triangle f , we denote its deformed edges given a warp field $\Phi : \mathbb{S}^2 \rightarrow \mathbb{S}^2$ by

$$\vec{e}_1(f; \Phi) = \Phi(\vec{x}_2) - \Phi(\vec{x}_1), \vec{e}_2(f; \Phi) = \Phi(\vec{x}_3) - \Phi(\vec{x}_1), \quad (8)$$

where $\vec{x}_1, \vec{x}_2, \vec{x}_3 \in \mathbb{S}^2$ are the associated vertices of f . We then compute the amount of the vertex-wise area deformation at location $\vec{x} \in \mathbb{S}^2$:

$$\Delta A(\vec{x}) = \frac{1}{6} \sum_{f \in \mathcal{N}_{\vec{x}}} \|\vec{e}_1(f; \Phi) \times \vec{e}_2(f; \Phi)\| - \|\vec{e}_1(f; I) \times \vec{e}_2(f; I)\|, \quad (9)$$

where I is an identity warp field, and $\mathcal{N}_{\vec{x}}$ is a set of the adjacent triangles at $\vec{x}$. We employ ΔA as a guidance to enhance the deformed geometric features G with attention based on scalar matrix multiplication [54]. Specifically, we use a feedforward network $\Psi : \mathbb{R}^C \times \mathbb{R} \rightarrow \mathbb{R}$ to generate vertex-wise attention that takes G and ΔA as its input. For each location $\vec{x} \in \mathbb{S}^2$, we have

$$\alpha(\vec{x}) = \Psi(G(\vec{x}), \Delta A(\vec{x})). \quad (10)$$

To ensure that $\alpha \in [0, 1]$, we further rescale α using the sigmoid function. This approach provides numerical stability by exponentiating and normalizing the logits, effectively preventing large values and potential numerical overflow issues:

$$a(\vec{x}) = \frac{1}{1 + e^{-\alpha(\vec{x})}}. \quad (11)$$

Finally, we generate the enhanced geometric features H by combining the refined weights a and the deformed geometric features G . To achieve this, we introduce another feedforward

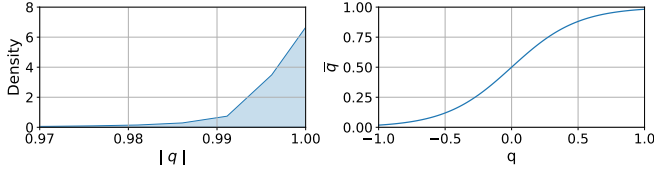


Fig. 6. Density function of the inner product between the triangle normal and centroid (*left*): the inner products are aggregated from all participants in the test set. Remarkably, 99.99% of these inner products are above 0.9. Sigmoid function at $\tau = 4$ (*right*): to control the inner products, the sigmoid function maps $|q|$ closer to 1. This adaptation effectively reduces the emphasis on the magnitude of q itself.

network $\rho : \mathbb{R}^C \rightarrow \mathbb{R}^D$, which encodes G into a higher-dimensional space D :

$$H(\vec{x}) = \rho(a(\vec{x}) \cdot G(\vec{x})). \quad (12)$$

The network ρ plays a crucial role in capturing and representing the enriched information contained in G , enabling the generation of enhanced geometric features H .

D. Sulcal Labeling Network

In this section, we discuss the design of $\mathcal{F}$. We assume that our labeling task is constrained by a limited number of manually annotated data and that the data are potentially misaligned. While rotational data augmentation can be employed to increase the quantity, it cannot sufficiently cover all possible data orientations of sulcal labels. Therefore, we utilize SPHARM-Net [44] to design $\mathcal{F}$, which supports rotational equivariance in spherical convolution. In each layer of SPHARM-Net, the input features H are converted into their spectral components via SHT at degree of l and order of m :

$$\hat{H}_{lm} = \int_{\mathbb{S}^2} H(\theta, \phi) Y_{lm}(\theta, \phi) d\theta d\phi. \quad (13)$$

The spherical convolution is then performed on the harmonic coefficients with a learnable zonal (isotropic) filter $\hat{K}_l \in \mathbb{R}$ for degree of l . The convoluted coefficients can be obtained by

$$h_{lm} = 2\pi \sqrt{\frac{4\pi}{2l+1}} \hat{H}_{lm} \hat{K}_l. \quad (14)$$

The resulting convolution preserves rotational equivariance because the norm of the harmonic coefficients remains unchanged in the spectral domain regardless of rotation [44]. The coefficients are converted into spatial signal via ISHT:

$$\bar{H}(\theta, \phi) = \sum_{l=0}^{\bar{B}} \sum_{m=-l}^l h_{lm} Y_{lm}(\theta, \phi) + \gamma H(\theta, \phi), \quad (15)$$

where $\bar{B}$ is spherical harmonic bandwidth, and γ is a learnable parameter to ensure that the convolution preserves detailed geometry by maintaining high-frequency components [44]. It is noteworthy that the convolutional filters in SPHARM-Net have limited expressive power due to their isotropic weighting. Nevertheless, such a limited representation can be overcome in our approach since the original input features are deformed by the proposed deformation module, which eases sulcal labeling even with isotropic filters. Once $\bar{H}$ is determined by $\mathcal{F}$, we

apply the inverse warp field Φ^{-1} to $\bar{H}$ because $\bar{H}$ is obtained from the warped sphere. Finally, we compute vertex-wise probability p per ROI via the softmax function of $\bar{H} \circ \Phi^{-1}$.

E. Losses

1) Self-Intersection Loss: To preserve an invertible smooth mapping, it is necessary to regularize the warp field derived from the feature deformation module. In [51], the regularization term is designed for spherical registration to maintain the warp field as isometric as possible to reduce warp distortion. However, such a constraint is too strong to effectively warp variable sulcal patterns into easily learnable forms in sulcal labeling. To address this issue, we introduce a novel regularization term that handles self-intersection of the warped sphere granting it greater flexibility. This regularization term can be considered the minimum requirement for a valid spherical warp. To this end, we take into account the disparity between triangle centroid $\vec{c}$ and normal $\vec{n}$ because $\vec{c}$ is always an outward vector and $\vec{n}$ determines the triangle orientation. For each triangle, let $\vec{x}_1, \vec{x}_2, \vec{x}_3$ be the associated vertices, respectively. Non-normalized centroid and normal vectors can be calculated by $\vec{c} = (\vec{x}_1 + \vec{x}_2 + \vec{x}_3)/3$ and $\vec{n} = (\vec{x}_2 - \vec{x}_1) \times (\vec{x}_3 - \vec{x}_2)$. Self-intersection is then assessed by their inner product:

$$q = \left\langle \frac{\vec{c}}{\|\vec{c}\|}, \frac{\vec{n}}{\|\vec{n}\|} \right\rangle. \quad (16)$$

Once again, our primary focus is on the inward/outward test of $\vec{n}$. Unfortunately, since $\vec{c}$ and $\vec{n}$ are not always parallel, their inner product q can be any real number in $[-1, 1]$. This can introduce some uncertainty when aggregating q . Therefore, we employ a differentiable sigmoid function to more strictly constrain q to be closer to 1 and 0 for the respective positive and negative values:

$$\bar{q} = \frac{1}{1 + e^{-\tau \cdot q_i}}, \quad (17)$$

where τ determines the slope. In practice, we have observed that nearly 99.99% of q lie above 0.9 (see Fig. 6). In icosahedral subdivision of 6, the observation suggests that the number of $q < 0.9$ is approximately 4 out of 81,920 triangles. To bring q closer to 1, we conservatively set τ to 4, which makes the slope of sigmoid function steeper. Because the sigmoid function is non-negative, we finally formulate the proposed self-intersection loss to be bounded in $[0, 1]$ as follows.

$$\mathcal{L}_{si} = 1 - \frac{1}{M} \sum_{i=1}^M \bar{q}_i, \quad (18)$$

where M is the number of triangles of spherical tessellation.

2) Sulcal Labeling Loss: As proposed in [23], we use cross-entropy to minimize the error between the ROI-wise probability p and its manual labels z :

$$\mathcal{L}_{label} = - \sum_{i=1}^L \log(p_i | z_i). \quad (19)$$

Finally, we evaluate the two terms in Eqs. (18) and (19) by a linear combination:

$$\mathcal{L} = \mathcal{L}_{si} + \beta \mathcal{L}_{label}, \quad (20)$$

where $\beta \in \mathbb{R}^+$ is a positive weighting factor used to balance the two loss terms.

IV. EXPERIMENTAL SETUP

A. Imaging Data

1) *HCP Dataset*: We used a subset of the Human Connectome Project (HCP) [55]. We randomly selected participants from the HCP for a total of 72 individuals under the following selection criteria. We chose an age range of 22 to 36 years (mean of 29.06 and standard deviation of 3.59) to cover almost the entire spectrum of the HCP dataset, maintaining a balanced gender ratio (36 females and 36 males). T1-weighted anatomical scans (voxel size of $0.8 \times 0.8 \times 0.8 \text{ mm}^3$) were acquired in their native space from the HCP database, along with outputs from the modified FreeSurfer pipeline [56]. The details on image acquisition parameters and image processing can be found in [56].

2) *NORA Dataset*: To validate the generalizability of the proposed method, we randomly selected 60 subjects (27 females and 33 males, aged between 6 and 18 years) from the Neurodevelopment of Reasoning Ability (NORA) project [57], [58]. The brain scans were obtained using a Siemens 3T Trio system at the University of California Berkeley Brain Imaging Center, employing high-resolution T1-weighted MPRAGE scans (TR=2,300 ms, TE=2.98 ms, voxel size of $1.0 \times 1.0 \times 1.0 \text{ mm}^3$). We employed a standard FreeSurfer pipeline [59] and then carefully checked each generated cortical surface for any segmentation inaccuracies, correcting them manually as needed.

3) *Sulcal Definitions*: We used the following labeling protocol for this study. First, screenshots of pial and inflated surfaces manually labeled by FreeSurfer's annotation tool were taken for each subject. Second, individual raters evaluated the labeled sulci using text and drawing tools in the screenshots. Third, definitions were refined and finalized through discussions regarding any discrepancies. Following this protocol, we defined 3 large (*i.e.*, primary/secondary), consistent sulci: marginal ramus of the cingulate sulcus (mcgs), parieto-occipital sulcus (pos), and the splenial sulcus (spl). We also considered 5 smaller and more variable (*i.e.*, putative tertiary) sulci for all participants: the inframarginal sulcus (ifrms), the precuneal limiting sulcus (preuls), and the posterior (preus-p), intermediate (preus-i), and anterior (preus-a) precuneal sulci [7], [25], [60]; see Table I for the nomenclature of cortical sulci. We used the left hemisphere for our study.

4) *Cortical Geometry*: The mean curvature of the cortical surfaces is widely used in cortical surface processing because it is mathematically well-defined and reasonably captures the overall cortical folding patterns. Nevertheless, its high locality could yield local optima of the optimization. Therefore, we used the following geometric features available in the FreeSurfer package [61]: mean curvature of the inflated surface (*inflated.H*), average convexity (*sulc*), and mean curvature of the white-matter surface (*curv*). This feature selection was inspired by the multi-scale geometric representations commonly utilized in spherical registration techniques [62]–[64]. The multi-scale approach generally gives a complementary shape representation at each scale from a coarse-to-fine perspective.

TABLE I
SULCAL DEFINITIONS OF POSTERIOR MEDIAL CORTEX.

Type	Acronym	Region
large, consistent	mcgs	marginal ramus of the cingulate sulcus
	pos	parieto-occipital sulcus
	spls	splenial sulcus
small, variable	ifrms	inframarginal sulcus
	preuls	precuneal-limiting sulcus
	preus-p	posterior precuneal sulcus
	preus-i	intermediate precuneal sulcus
	preus-a	anterior precuneal sulcus

B. Implementation Details

We re-tessellated individual spheres by icosahedral subdivision with 40,962 sampling points [65]. For the deformation module and sulcal labeling network, we tweaked their entry channels C_{deform} and C_{sulc} to 64 and 128, respectively. We set the harmonic bandwidth $\bar{B} = 120$ for the SPHARM-Net instances in the deformation module and the sulcal labeling network. We set the bandwidth $B = 20$ for the warp field estimation in Eq. (6) and $k = 6$ for the numerical integration in Eq. (7). The training was performed in an end-to-end manner with the Adam optimizer [66] at an initial learning rate of 0.01 with decay by a factor of 0.1 if no improvement is made in two consecutive epochs. We set $\beta = 1/10$ for the total loss in Eq. (20). We used PyTorch for the backend processing and customized CUDA kernels for the spherical tessellation of warped spheres.

C. Baseline Methods

Due to the absence of existing methods that were evaluated on the annotation protocol defined in this study, we used four generic baseline methods for comparison: a multi-atlas approach [23] and spherical CNNs [20], [44], [67] for cortical surface processing. For a fair comparison, we computed an optimal rigid alignment for each individual participant within the training samples to avoid any potential performance degradation caused by data misalignment.

For the multi-atlas approach, we followed a leave-one-out majority voting scheme [68]. Specifically, we used FreeSurfer's spherical registration [62] to establish shape correspondence with *fsaverage* provided by the FreeSurfer package. After surface registration, we determined the final label for each vertex by the most frequently observed label from all registered spheres except for the sphere being evaluated. For all the CNN methods, including ours, we used the three geometric features and non-rigid spherical data augmentation to alleviate data scarcity [23]. Given that the baseline methods were specifically designed for cortical surface parcellation, we fine-tuned them to maximize their sulcal labeling performance. Furthermore, to ensure a fair comparison, we matched the network parameters across all baseline CNN methods for extensive evaluation. For Spherical U-Net [20], we tuned the entry channel of 32 with icosahedral subdivision of 4 (1.67M parameters), the entry channel of 32 with icosahedral subdivision of 5 (5.12M parameters), and the entry channel of 64 with icosahedral subdivision of 5 (26.86M parameters). For UG-SCNN [67], we tuned the entry channel

of 32, 64, and 128, yielding the parameters of 1.70M, 5.18M, and 20.67M, respectively. For SPHARM-Net [44], we tuned the entry channel of 64, 128, and 256, yielding the parameters of 1.10M, 4.31M, and 17.01M, respectively.

D. Evaluation Metrics and Training Strategies

We evaluated the performance in Dice score (whole cortex and individual ROIs). We conducted paired t -tests to demonstrate statistical significance against the baseline methods. In ROI-wise analysis, the statistical significance was reported after multi-comparison correction among the 8 sulci using false discovery rate (FDR) [69] at $q = 0.05$. We employed 5-fold cross-validation, where the original dataset was split into 5 partitions with 60% for training (3 partitions), 20% each for validation and test (1 partition). For each fold, we circulated the partitions to achieve full validation. In the test phase, we used the learned model parameters at the peak accuracy in the validation set. The vertex-wise discriminant in the test set was then transferred to the original surface tessellation to determine final labels [23]. All experiments were run on an Intel Xeon 6248R and an NVIDIA RTX 3090 GPU device.

V. RESULTS

A. Hyperparameter Choice

We justify our choice of the five hyperparameters: channel size C_{sulc} of the sulcal labeling network, channel size C_{deform} of the deformation module, harmonic bandwidth B , numerical integration step k , and loss balance β . For each hyperparameter, the other components were fixed to the default settings described in Section IV-B. We varied C_{sulc} , C_{deform} , B , k , and β in $\{16, 32, 64, 128, 256\}$, $\{8, 16, 32, 64, 128\}$, $\{10, 15, 20, 25, 30\}$, $\{0, 2, 4, 6, 8\}$, and $\{1/25, 1/20, 1/15, 1/10, 1/5\}$, respectively. Fig. 7 shows a trend of each hyperparameter.

1) *Channel Size of Sulcal Labeling Network* : The overall performance is improved in Dice score as C_{sulc} increases. We observed a significant improvement in the Dice score up to $C_{sulc} = 64$ followed by a gradual convergence, likely because of the limited capacity of the extracted features to provide distinct characteristics for sulcal labeling beyond $C_{sulc} = 64$. In this study, we chose $C_{sulc} = 128$ to achieve a balance between performance and computational efficiency.

2) *Channel size of Deformation Module* : Similar to $C_{sulc} = 64$, we observed performance improvement as C_{deform} increases. There is a notable improvement at $C_{deform} = 32$, followed by convergence. This trend likely stems from the same reason observed in C_{sulc} . We chose $C_{deform} = 64$ to achieve a balance between performance and computational efficiency.

3) *Harmonic bandwidth* : As the harmonic bandwidth increases, the feature deformation module becomes more flexible, leading to an improved Dice score. We observed this trend beginning to converge at $B = 20$ likely because the feature deformation adequately captures sulcal variability. Although increasing B provides a higher degree of freedom, we observed only marginal improvements after $B = 20$ due to the saturation of improvements in the spherical warp.

TABLE II

ABLATION STUDY ON THE PROPOSED FEATURE DEFORMATION MODULE.

Deformation	Target	Dice (%)	p value
$\times$	-	62.92 ± 14.32	8.9×10^{-3}
	$\checkmark$	63.01 ± 14.12	8.1×10^{-3}
$\checkmark$	$\times$	67.26 ± 13.28	-

TABLE III

ABLATION STUDY ON THE PROPOSED GUIDED-ATTENTION MODULE AND COMPARISON TO ITS ALTERNATIVE DESIGN CHOICES.

Guidance	Design Choice	Dice (%)	p value
$\times$	-	65.87 ± 13.45	7.8×10^{-3}
	self-attention	66.01 ± 14.82	1.2×10^{-3}
	concat.	65.96 ± 13.29	7.1×10^{-3}
$\checkmark$	concat. and linear	66.38 ± 14.74	1.7×10^{-3}
	guided-attention	67.26 ± 13.28	-

4) *Integration step* : The accuracy of the numerical integration improves with increasing k . Only negligible differences are noted after $k = 6$. We selected $k = 6$ to achieve an optimal trade-off between performance and computational efficiency. We found no self-intersection after $k = 6$.

5) *Loss Balance* : The significance of the feature deformation module in the loss function is determined by β . As β increases, the Dice score improves, eventually converging at $\beta = 1/10$. Similar to B , β restricts a degree of freedom in the spherical warp. Although no self-intersection was found in our experiments, we conservatively chose $\beta = 1/10$ to achieve a balance between performance and numerical stability.

B. Ablation Study

To validate our design choices, we conducted a quantitative comparison by disabling individual components and evaluating their impact. For all experiments, we used the default parameter settings as described in the implementation details.

1) *Feature Deformation Module*: To assess the effectiveness of the proposed feature deformation module, we conducted an ablation study by excluding the module. This results in the sulcal labeling network alone, which is equivalent to SPHARM-Net. Table II summarizes the effectiveness of the proposed module in Dice score. Clearly, the feature deformation module facilitates the sulcal labeling process. To verify the necessity of an explicit target for the deformation module, the proposed module was trained to match the target geometry ($fs_{average}$), similar to spherical registration. As expected, there was no improvement because anatomical variability is not well represented in the target geometry, which rather constrains the flexibility of the spherical warp (see Table II). As also observed in Fig. 2, the average sulcal patterns indeed diminish as more subjects are aligned to the target.

2) *Guided-Attention Module*: To investigate the effectiveness of the guided-attention module, we performed an ablation study by excluding the module. Since the importance of labeled regions is stressed by the proposed module, the overall Dice score can be improved (see Fig. 8). We further considered the following design choices to effectively utilize the extent of deformation for a guidance: (1) *self-attention*: leveraging the input geometric features themselves as self-guidance through

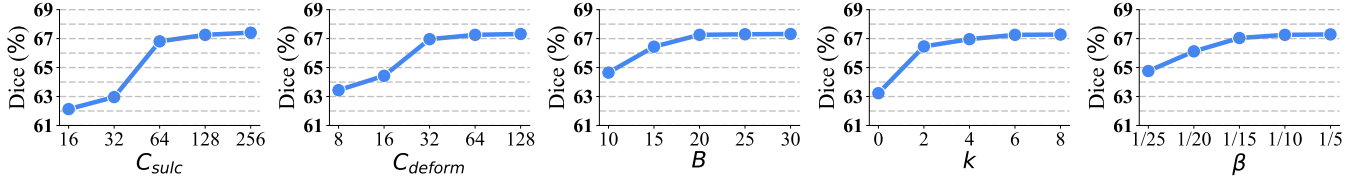


Fig. 7. Hyperparameter tuning. Channel size C_{sulc} of the sulcal labeling network and channel size C_{deform} of the deformation module: the Dice score is improved as C_{sulc} and C_{deform} increase. Harmonic bandwidth B : the Dice score is improved as B increases. Numerical integration step k : the Dice score is improved and no self-intersection is found after $k = 6$. Loss balance β : the Dice score is improved as β increases.

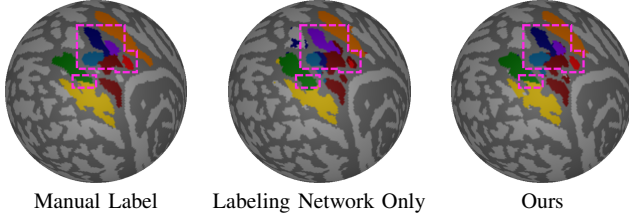


Fig. 8. Qualitative comparison between the sulcal labeling network only (equivalent to SPHARM-Net) and the proposed method with full functionality including feature deformation and guided-attention modules. The proposed method simplifies the boundaries between putative tertiary sulci, which reduces sulcal variability.

TABLE IV

ABLATION STUDY ON THE PROPOSED SELF-INTERSECTION LOSS.

Self-Intersection	Dice (%)	p value
✗	64.35 ± 14.11	2.0×10^{-4}
✓	67.26 ± 13.28	-

a scalar matrix multiplication-based attention mechanism [54], (2) *concat.*: simple concatenation of geometric features and the extent of spherical deformation, (3) *concat. and linear.*: passing concatenated features through a feedforward network, and (4) *guided-attention*: using the extent of spherical deformation with a scalar matrix multiplication-based attention mechanism (see Sec. III-C). As summarized in Table III, the proposed method outperforms alternative design choices. Clearly, the inclusion of an attention mechanism outperforms alternative approaches in sulcal labeling accuracy. This is because our approach prioritizes sulcal regions by effectively encoding both the geometric features and the extent of deformation induced by the proposed feature deformation module.

3) *Self-Intersection Loss*: To validate the effectiveness of the proposed self-intersection loss, we conducted a comparison with the regularization loss used for spherical registration in our previous work [51]. As shown in Table IV, there is a performance gap between the two regularization terms. This is mainly because the conventional regularization in spherical registration is designed to prevent the deformed surface from excessively deviating from the initial grid. This observation suggests that the conventional regularization in spherical registration is limited in its ability to fully exploit the advantages of feature deformation, whereas the proposed regularization achieves this goal effectively.

4) *Feature Selection*: To investigate the importance of each input geometric feature, we incrementally added coarse-to-fine grained features describing cortical geometry: *inflated.H*,

TABLE V

ABLATION STUDY ON THE SELECTION OF INPUT GEOMETRIC FEATURES.

Input Features	Dice (%)	p value
<i>inflated.H</i>	62.43 ± 15.04	2.2×10^{-5}
<i>sulc</i>	63.46 ± 14.24	4.4×10^{-4}
<i>curv</i>	64.97 ± 14.22	2.9×10^{-4}
<i>inflated.H</i> + <i>sulc</i>	65.87 ± 13.77	1.7×10^{-3}
<i>inflated.H</i> + <i>sulc</i> + <i>curv</i>	67.26 ± 13.28	-

sulc, and *curv*. In Table V, significant performance improvement is observed upon incrementally adding the three geometric features. The proposed method achieves an overall performance of 62.43% of Dice score when trained using only the coarse feature *inflated.H* that captures global but limited local geometry. Incorporating *sulc* as an additional feature to *inflated.H* leads to a significant performance boost of 3.4%p. This improvement is observed because the finer-grained cortical geometry (*sulc*) offers a complementary level of detail to the coarse representation provided by *inflated.H*. After incorporating all three features, we observed a 1.4%p improvement in performance. This suggests that the inclusion of both coarse- and fine-grained features is beneficial for sulcal labeling as they exhibit complementary characteristics.

C. Feature Deformation

To understand how the proposed feature deformation module effectively warps the input geometry, we investigated the morphological changes of ROIs before and after warping. For quantitative analysis, we calculated the coefficient of variation for the area of each ROI. This quantification captures the area variation of each ROI by normalizing the area variation independent of class-wise area differences. A lower quantity indicates a more tightly constrained data distribution. As shown in Fig. 9, the area variation is reduced across all ROIs. Meanwhile, the area of ROIs increases by up to 1.6 times for the larger primary/secondary regions and up to 4.5 times for the smaller putative tertiary regions. This result indicates that the proposed module effectively alleviates sulcal variability in terms of their size and facilitates sulcal labeling by increasing the area size. Additionally, unlabeled regions are inevitably reduced in size as sulcal ROIs expand. Quantitatively, the portion of the unlabeled area on the sphere decreases from 95.22% to 92.51%. Nonetheless, this reduction does not necessarily imply an adverse effect on identifying the unlabeled regions because they are ultimately grouped into a single region to be labeled as a background. Specifically, when neither the feature deformation nor guided-attention modules were applied, the

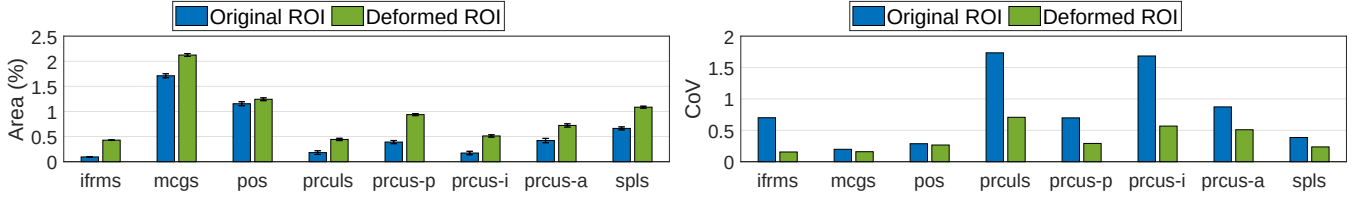


Fig. 9. Area change before and after feature deformation (*left*): the proposed feature deformation module scales up all the regions while reducing the deviation of each ROI. Encouragingly, smaller putative tertiary sulci are more expanded than larger primary/secondary sulci to improve sulcal labeling. Coefficient of variation (CoV) for area of each ROI before and after feature deformation (*right*): a reduced coefficient of variation suggests that the proposed module effectively reduces sulcal variability by simplifying the ROI boundaries.

Dice score is 99.54% for the grouped region, whereas the proposed method achieves a Dice score of 99.96%.

D. Comparisons to Baseline Methods

1) *Overall Dice Score*: We varied the entry channel size of the proposed method for a fair comparison to the baseline methods: 32, 64, and 128, yielding 880K, 3.35M, and 7.82M, respectively. Table VI shows that all the methods tend to achieve better performance as their model size becomes larger, where the proposed method outperforms all the baseline methods. Notably, our smallest model (880K) achieves even better performance compared to the larger models of the baseline ($p < 0.05$).

2) *ROI-wise Dice Score*: In the ROI-wise analysis, we compared each baseline method as summarized in Table VI. Clearly, the proposed method outperforms the baseline methods across all the regions while no performance degradation is observed as shown in Fig. 10. Particularly in the smaller sulci such as ifrms, prcus-p, prcus-i, and prcus-a, the proposed method exhibits remarkable performance improvements, with a Dice score increase of more than 5%p compared to the baseline methods. Notably, the proposed method achieves a substantial performance gain of approximately 10%p in the smallest region (ifrms). These findings highlight that the proposed method is useful particularly in labeling putative tertiary sulci. Fig. 11 shows qualitative comparison results around the average performance. The qualitative analysis shows consistent improved labeling performance particularly in putative tertiary sulci, as also evidenced by the ROI-wise Dice score in Fig. 10.

E. Generalizability

To highlight that the proposed method works when applied to a different age range, we chose pediatric samples from the NORA dataset. The same hyperparameter settings of all the methods were used as configured in the experiments on the HCP dataset. It is noteworthy that the experiment is particularly challenging due to the inherent characteristics of pediatric data: (1) relatively greater anatomical variability [70], [71], particularly across a large age range (6–18 years), and (2) a reduction in sample size from 72 to 60 (approximately 83% of the HCP dataset). Nevertheless, a two-sample t -test reveals no significant difference in Dice score between the two age groups ($p = 0.306$). The proposed method outperforms the baseline methods similar to those observed in the HCP dataset

TABLE VI

OVERALL DICE SCORE FOR COMPARISON TO BASELINE METHODS.

Method	Dice (%)	p value
Multi-atlas [68]	47.01 $\pm$ 9.15	1.1×10^{-21}
Spherical U-Net (1.67M) [20]	61.44 $\pm$ 15.21	8.9×10^{-6}
Spherical U-Net (5.21M) [20]	62.03 $\pm$ 14.85	1.7×10^{-5}
Spherical U-Net (26.86M) [20]	62.50 $\pm$ 14.95	1.9×10^{-5}
UG-SCNN (1.70M) [67]	58.93 $\pm$ 15.45	1.8×10^{-10}
UG-SCNN (5.18M) [67]	61.72 $\pm$ 14.33	2.0×10^{-6}
UG-SCNN (20.67M) [67]	62.68 $\pm$ 15.24	3.4×10^{-5}
SPHARM-Net (1.10M) [44]	62.88 $\pm$ 14.51	8.3×10^{-5}
SPHARM-Net (4.31M) [44]	62.92 $\pm$ 14.32	3.0×10^{-5}
SPHARM-Net (17.01M) [44]	63.09 $\pm$ 14.36	1.6×10^{-4}
Ours (880K)	66.76 $\pm$ 14.11	0.57
Ours (3.35M)	66.81 $\pm$ 13.64	0.59
Ours (7.82M)	67.26 $\pm$ 13.28	-

TABLE VII

OVERALL DICE SCORE ACROSS PEDIATRIC PARTICIPANTS.

Method	Dice (%)	p value
Multi-atlas [68]	45.98 $\pm$ 7.96	2.4×10^{-20}
Spherical U-Net (26.86M) [20]	62.76 $\pm$ 10.81	1.7×10^{-5}
UG-SCNN (20.67M) [67]	62.88 $\pm$ 11.48	2.0×10^{-5}
SPHARM-Net 17.01M) [44]	63.12 $\pm$ 11.76	3.0×10^{-5}
Ours (7.82M)	67.12 $\pm$ 10.58	-

(see Table VII). Therefore, these findings provide evidence for the generalizability of the proposed method across a broad age range (6–36).

VI. DISCUSSION

A. Sulcal Variability

In Fig. 1, significant structural variations across individuals can be observed, especially in the smaller putative tertiary sulci. This variability poses a major challenge in automatic sulcal labeling. In this study, we introduced a learning-based framework that effectively scales up ROIs and mitigates sulcal variability. Here, the increased area corresponds to an increased number of sampling points in the spherical tessellation, particularly in small ROIs. This results in an improved sulcal labeling performance as the proposed method gains more informative data points to accurately identify the sulcal regions. In addition, the limited flexibility in representation (*i.e.*, isotropic filters) in the sulcal labeling network driven by SPHARM-Net is effectively addressed through the proposed deformation module, while maintaining rotational equivariance (see Table II). Overall, the proposed method can accurately

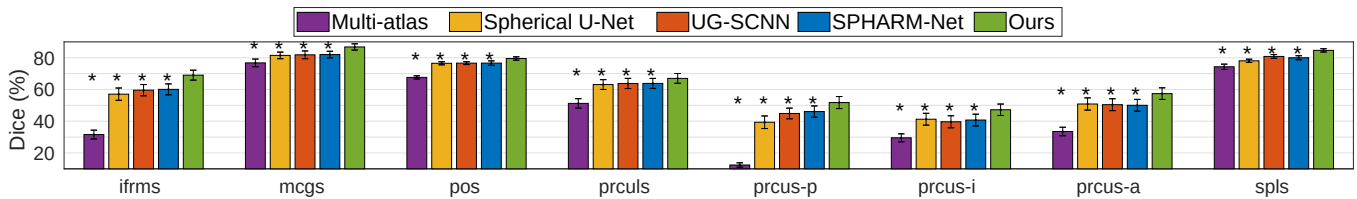


Fig. 10. Region-wise Dice score evaluation. The statistical significance is reported after multi-comparison correction among the 8 sulci. The proposed method exhibits significantly enhanced performance across all regions, with a particularly pronounced performance gap observed in the smaller putative tertiary sulci. Legend: standard errors (hat); significant improvement than the baseline methods (*).

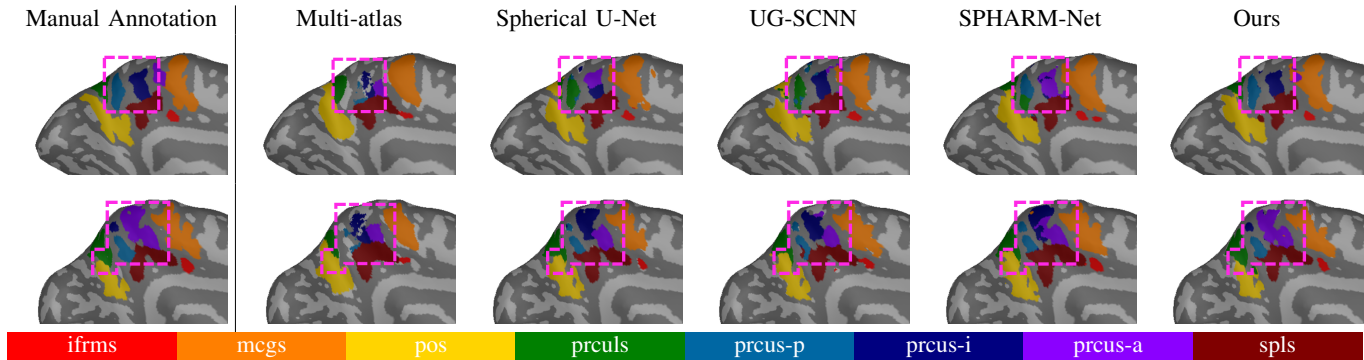


Fig. 11. Visual inspection on two example participants around the average Dice score. The proposed method outperforms the baseline methods, particularly in small ROIs (prculs, prcus-p, prcus-i, and prcus-a), where the boundaries between sulci are well recognized. Inflated surfaces are used for better visualization. The pink box highlights improved performance.

label ROIs by simplifying sulcal patterns on the unit sphere. Notably, we observed a greater improvement in labeling of putative tertiary sulci that exhibit larger deviations (see Fig. 10).

B. Class Imbalance Issue

Our labeling task involves a class imbalance problem, given large unlabeled regions and relatively small ROIs. The prevalence of unlabeled regions can obscure the learning of sulcal ROIs since these regions mostly consist of less informative sulcal patterns, potentially causing confusion during the learning process. To address this issue, we introduced an attention mechanism to focus on important regions while suppressing less informative regions. Specifically, we introduced warp field-based guidance under the assumption that regions with a greater expansion hold higher importance. As a result, we significantly reduced the computational demands and efficiently encoded the importance of sulcal ROIs. Moreover, the performance improvement in sulcal ROIs also translates to enhanced performance in unlabeled regions. This addressed the class imbalance issue in our labeling task. Meanwhile, a self-attention mechanism like transformers [72], [73] is a popular choice to address the class imbalance. This approach learns the importance implicitly from the input data, requiring quadratic computational complexity. Unfortunately, this approach is currently implausible for our task due to the limited availability of annotated data. In future work, we will investigate an advanced attention-based approach for potential improvement in handling the class imbalance issue.

C. Generalizability

The cortical surfaces include large primary/secondary and small putative tertiary sulci. To investigate the generalizability

of the proposed method, we used different age groups in this study. Despite substantial anatomical variability, we found no significant difference between different age groups in Dice score. Although we have included pediatric and adult participants in this study, spanning ages 6–36, expanding the inclusion criteria to encompass a broader age range will further support the generalizability of the proposed method. Also, given that the current study solely targets the PMC regions, exploring other cortical regions will offer additional support for the generalizability of the proposed method, which we leave for future work.

D. Sulcal Labeling Performance

For each putative tertiary sulcus, our method exhibited performance improvements of over 5%p. Given the greater sulcal variability and the smaller size of these ROIs, this can be regarded as a significant improvement in performance. Despite the significant advancement achieved by the proposed method, the absolute performance in putative tertiary sulci still remains lower than that in the larger primary/secondary sulci, primarily due to the inherent substantial anatomical variability found in the smaller tertiary sulci. To narrow the performance gap between primary/secondary and putative tertiary sulci, several potential approaches can be explored. First, a hierarchical architecture design could be employed, integrating multi-resolution features [47], [48] to enable effective representation across a range of scales, from small to large regions. Second, auxiliary networks could be incorporated to leverage the context information [45], [46] from primary/secondary sulcal regions as priors for tertiary sulci holds promise. Also, a similar strategy could be used for tertiary sulci, allowing for a

hierarchical subdivision of integrated smaller sulci (e.g., *prcus* first and then *prcus-p*, *prcus-i*, and *prcus-a*). These approaches have the potential to enhance sulcal labeling across various levels of anatomical complexity, which we leave for future research. Finally, alternative cortical surface measures such as functional gradient density maps or myelination patterns may improve sulcal labeling. We plan to investigate these measures in our future work to explore their impact on labeling.

VII. CONCLUSION

In this study, we introduced a learning framework for sulcal labeling in PMC. We aimed to address the substantial anatomical variability, the class-imbalance problem, and the limited availability of manually annotated data. We proposed an end-to-end learning framework utilizing a spherical CNN. The proposed method warps input features to facilitate sulcal labeling while mitigating the anatomical variability. Furthermore, we introduced a guided-attention mechanism that focuses on regions to be labeled while suppressing irrelevant information in unlabeled regions. We validated the proposed method on the pediatric and adult cohorts. We found that the proposed method outperforms the baseline methods across children and adults, particularly in small, variable sulci.

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